Regression Analysis of US Telecom Reliability, Availability, and Maintenance (RAM) and Customer Experience (CX) Variables and Their Effects on EBITDA

Jeffrey D. Hill

Department of Business, Edgewood University

BUS 920 – Quantitative Methods I

Professor Kenneth A. Kavajecz

Abstract

This study examines the impact of reliability, availability, and maintenance (RAM) and customer experience (CX) variables on the financial performance of U.S. telecommunications companies, measured by adjusted EBITDA. Data was collected from the FCC's Measuring Broadband America program (2018–2022), ACSI telecommunications reports (2018–2022), and associated SEC 10-K filings. A multiple linear regression model was applied to twelve independent variables. These variables include jitter, latency, packet loss, web page load times, download and upload speeds, compliance with advertised speeds, and customer satisfaction indices for TV, internet, landline, and video on demand services. The analysis found that most reliability and speed measures were not significant predictors of financial outcomes. The strongest significance came from customer perception of receiving the advertised upload speed. This suggested that perceived value is more closely tied to EBITDA than raw technical performance. The null hypothesis was rejected in favor of a mixed outcome, where some RAM and CX factors influence profitability but neither group alone provides full explanatory power. The findings emphasize the importance of meeting customer expectations and call for further research on service-type segmentation, maintenance variables, and churn data.

Introduction

Telecommunications is a stable investment sector even during recessions due to being a vital service and exhibiting characteristics of both income and growth stocks (Beers, 2025). Telecommunications is also a highly competitive sector requiring companies to keep up with speed, data connectivity, higher resolutions, and multimedia services (Beers, 2025). According to Wierowski, telecommunication companies need to focus on "'five Cs': convergence, customer experience, commercial innovation, cost excellence, and complexity management (2025). Commoditization is impacting financial performance in the industry (Wierowski, 2025). To keep up with these performance characteristics companies require significant capital and must be large enough to produce significant cash flow (Beers, 2025). This study is to see the effect of notable variables tracked by the Federal Communications Commission (FCC) which is the governmental oversight organization for telecommunications and combined with indexes provided by the American Customer Services Index (ACSI) annual reports.

Question

The question is do the variables: jitter, latency, packet loss, web page load times, download speeds, meeting advertised download speeds, upload speeds, meeting advertised upload speeds, TV service index, Internet service index, landline service index, and Video on Demand (VOD) service index, explain changes in the earnings before interest, taxes, depreciation, and amortization (EBITDA) of telecommunications companies. By identifying and reducing non-significant variables of these combined technical and customer experience factors telecommunications can focus on the five Cs.

Importance

Determining which variables in reliability, availability, and maintenance along with the customers perspective of experience affect the bottom line can assist telecommunication companies in strategizing efforts to improve service delivery, perception, or experience. This

will enhance efforts for a company to stand out in a highly competitive industry undergoing commoditization.

Data

Independent Variables FCC

Variables identified in *Appendix: Glossary of Technical Terms* were collected from the FCC statistical reports from 2018 to 2023 of the Measuring Broadband America Program (FCC OET, 2018-2023). The Measuring Broadband America Program was a multi-year initiative with Samknows an FCC contractor, to create a standard methodology for measuring Internet performance globally (FCC OET, n.d.). In 2023 Samknows was purchased by Cisco Systems and is no longer an FCC partner. New sources are needed for future studies (FCC OET, n.d.).

Independent Variables ACSI

ACSI provides national cross-industry benchmarks as a measure of customer satisfaction within the U.S. Economy (ACSI LLC, n.d.). Indexes were collected from ACSI Telecommunications Index reports from 2018 to 2022 for companies listed in the FCC tests (ACSI LCC, 2018-2022). These companies include: CenturyLink, Inc.; Lumen Technologies, LLC.;, Frontier Communications Parent, LLC.; Charter Communications, Inc. (Spectrum), Comcast Corporation; Altice USA, Inc. (Optimum); Verizon Communications, Inc. Index values were collected for Television Service, Internet Service Providers, Fixed-Line Telephone, and Video on Demand (VOD). These indices are the components of triple play service.

Dependent Variable Security Exchange Commission

The Security Exchange Commission (SEC) provides financial reporting from its EDGAR database. Normally for telecommunications, ARPU or Average Revenue per User is the preferred measure of financial well-being. ARPU is not regularly available in the EDGAR database. Some companies provide ARPU in exhibits or on their websites, but not all provide and none for all years in the FCC reports. To match the independent variables to financial success of the

telecommunications companies adjusted EBITDA was chosen. This is a measure of a company's earnings and normalizes the result to account for anomalies.

Hypothesis

Financial success of a telecommunications business is determined by several factors. The focus of this study is the impact of Reliability, Availability, Maintenance (RAM) matched with reported Customer Experience (CX) on a company's bottom line. Reliability can be determined using the variables of jitter, latency, and webpage load time. Availability is determined by using download speeds, upload speeds, and their associated advertised amounts. Customer experience is determined by using ACSI indices.

This study used multi-linear regression over the twelve variables collected from the years 2018 to 2022. Microsoft's Excel data analysis tools were used to perform the regression. Results will be analyzed to determine whether they fit the following hypotheses:

H_0: RAM and CX have no impact on Adjusted EBITDA. $\beta_i = 0$ for all independent variables.

H_1: RAM, but not CX has impact on Adjusted EBITDA. $\beta_i > 0$ & f-stat < 0.000001 with RAM t-stats $|t| > 2$ and CX t-stats $|t| < 2$.

H_2: CX, but not RAM, has an impact on Adjusted EBITDA. $\beta_i > 0$ & f-stat < 0.000001 with CX t-stats $|t| > 2$ and RAM t-stats $|t| < 2$.

H_3: RAM and CX have a significant impact on Adjusted EBITDA. $\beta_i > 0$ & f-stat < 0.000001 with RAM t-stats $|t| > 2$ and CX t-stats $|t| > 2$.

H_4: RAM and CX have an impact on Adjusted EBITDA, with some variables providing significance. $\beta_i > 0$ & f-stat < 0.000001 with some RAM or CX stats $|t| > 2$.

Analysis of Results

Reviewing the regression results we see the $\beta_i > 0$ for multiple variables indicating that we have enough evidence to reject the null hypothesis. Therefore, the variables selected have at

least some impacts on a telecommunication company's bottom line. Furthermore, the f-stat noted significance F, is 1.12601E-06 indicating that the overall is statistically significant. Multiple R of 0.9256 indicates there is a strong correlation with some of the predictor variables. The R squared value of 0.8568 indicates the model has strong explanatory power. The adjusted R squared value of 0.7787 indicates that one or more of the predictors selected do not provide meaningful results.

A review of each predictor provides more information on their individual significance. The first four predictors are tied to reliability. The intercept, β_0, provides no meaningful information. Jitter, β_1 value was 2844.11, had a t-stat of 1.09, and a p-value of 0.29. This indicates that jitter is a poor predictor of the bottom line. Latency RTT, β_2 value, was 270.89, had a t-stat of 0.98, and a p-value of 0.34. This indicates that latency is also a poor predictor. Packet loss, β_3 value was -762325.30, had a t-stat of 0.34, and a p-value of 0.73. The negative value of the coefficient suggests there is an inverse relationship with the bottom line. It makes sense that as more loss occurred the bottom line would go down. Interestingly though the t-stat and p-value indicate that packet loss does not have a significant impact. It can be questioned whether packet loss in significant amounts under more observations might show more impact since the data provided to the FCC could be considered competitive to be the best service and not what a user might experience during the worst service. Webpage load times, β_4 value was 2.27, had a t-stat of 0.75, and a p-value of 0.46. This indicates that this predictor is not significant in determining the bottom line. Furthermore, the predictor coefficient makes little sense with a positive value. Increasing web page load times should not produce an increase in the bottom line.

The next four predictors are tied to availability. Download speed, β_5, value was 21.41, had a t-stat value of 1.33, and p-value of 0.20. The t-stat value indicates some borderline significance. Download speed versus advertised download speed, β_6, value was -50271.25, had t-stat value 1.55, and p-value of 0.14. The t-stat value indicates borderline significance. Upload speed, β_7, value was 13.70, had a t-stat of 0.73, and a p-value of 0.48. These values indicate

that upload speeds have no real significance on bottom line. Upload speed versus advertised upload speed, β_8, value was 108587.24, had a t-stat value 3.83, and a p-value of 0.00092. These values indicate that having upload speeds as advertised is highly significant to the bottom line. It is understood from these four variables that the actual speeds available are less important than the perception of speed that was purchased. This makes sense that a customer purchases speed based on what they can afford or need and therefore the actual speeds matter little. But there is also the understanding that a customer not getting what they paid for has more significance.

The final four predictors relate to customer experience. ACSI TV Service index, β_9, value was -553.00, had a t-value 0.54, and a p-value 0.59. The negative value for beta made little sense. The quality index should increase positively with the bottom line. It must be understood that this can be an artifact of overlap by other variables. This may also be due to the low variance within the index values. More observations are needed. Nevertheless, its value as a predictor was insignificant. ACSI Internet Service index, β_{10}, value was 541.56, had a t-stat of 0.69, and a p-value of 0.49. It is another insignificant predictor. ACSI Land-line index, β_{11}, value was 1669.42, had a t-stat of 1.89, and a p-value of 0.49. It is interesting to note that this t-stat shows more significance than other t-stats from these indices. In telecommunications the number of landlines is declining, and mobile communications is increasing. This index may have had lower number of respondents with more consistent answers. Also note that landlines provide less correlation with other services in these indices. ACSI Video on Demand index, β_{12}, value was 530.37, had a t-value of 0.56, and a p-value of 0.58. This is another predictor that fails to show significance to the bottom line. It would be reasonable to re-run this experiment with the service infrastructure types such as DSL, cable, fiber, mobile, or satellite instead to see if those had any effect. This might help eliminate correlation between over-the-top services like TV, Internet, and Video on Demand.

Altogether we do see significance within the predictors selected towards a company's adjusted EBITDA. We can reject the null hypothesis that RAM and CX have no impact on the

bottom line. We choose our alternate hypothesis H_4 stating that some of the selected predictor variables of RAM and CX are statistically significant, but neither RAM, nor CX alone were statistically significant.

Conclusion

Implications

From the data and significance found in the regression the perception of value variables, upload and download speeds compared to advertised are good predictors in determining a company's adjusted EBITDA is significant. These variables were initially noted as availability variables but may need to be re-categorized as perception of value variables. Later it was discovered that ETSI provided a category called Quality of Service/Experience or QoS/E (1994).

Reflection

During this study, a lack of data for private companies was noticed. Some private companies were amongst the volunteers that participated in FCC testing. This allowed them free access to data about their own performance. Data made available to the public through the transparency policy was limited to the month of September of each of the years. According to the FCC the participants attributed 80% of the U.S. consumer market (FCC OET, n.d.). An issue arose trying to compare private companies and adjusted EBITDA from SEC filings. This led to eliminating privately owned companies and requiring more years of data to provide for the number of observations in this study. Another issue was trying to initially use ARPU and churn. ARPU is a measure best suited for financial health of telecommunications, but it is often not publicly available or found in quarterly or annual slides for each company. Had this study been prepared prior to those postings it could be set as a task to collect those quarterly results. Churn is another measure of customer satisfaction and stickiness. It is also another dependent variable used in papers. Churn can also be found in quarterly or annual slides for companies. Some sites have only the past 3 years, if any. For ideas for future study see *Appendix: Areas of Future Interest.*

Appendix: Table 1 - Regression Results

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.9256
R Square	0.8568
Adjusted R Square	0.7787
Standard Error	7163.2944
Observations	35

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	12	6753047072	562753922.7	10.9671	1.12601E-06
Residual	22	1128881301	51312786.42		
Total	34	7881928373			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-208068.50	42160.75	-4.94	0.00	-295504.55	-120632.45	-295504.55	-120632.45
2.23	2844.11	2597.64	1.09	0.29	-2543.07	8231.29	-2543.07	8231.29
35.34	270.89	275.28	0.98	0.34	-299.99	841.78	-299.99	841.78
0.0021	-762325.30	2220649.19	-0.34	0.73	-5367669.84	3843019.24	-5367669.84	3843019.24
2904.73	2.27	3.03	0.75	0.46	-4.01	8.56	-4.01	8.56
29.1	21.41	16.04	1.33	0.20	-11.86	54.67	-11.86	54.67
0.8794	-50271.25	32531.54	-1.55	0.14	-117737.53	17195.02	-117737.53	17195.02
2.98	13.70	18.89	0.73	0.48	-25.47	52.87	-25.47	52.87
0.9188	108587.24	28388.11	3.83	0.00	49713.90	167460.59	49713.90	167460.59
63	-553.00	1019.63	-0.54	0.59	-2667.58	1561.59	-2667.58	1561.59
60	541.56	779.71	0.69	0.49	-1075.46	2158.58	-1075.46	2158.58
65	1669.42	885.13	1.89	0.07	-166.23	3505.07	-166.23	3505.07
63	530.37	951.38	0.56	0.58	-1442.66	2503.41	-1442.66	2503.41

Appendix : Glossary of Technical Terms

Download sustained trimmed mean (Mbps) peak weighted mean. This is a measure of download speed measured in megabits per second at peak hours (FCC OET, n.d.).

Download sustained trimmed mean unweighted mean of percentage advertised. This is the ratio of the download speed compared to the advertised download speed converted to percentage.

Jitter. Jitter is the short-term variation in a timing signal from its ideal position in time (ITU TSS, 1996). In respect to internet traffic this is normally the variation from the mean in latency detected when using the ping command.

Latency. Latency, short for packet latency, is the round-trip time it takes a packet to reach its destination and return to its source (ITU TDB, 2020).

Packet loss. Packet loss is simply the loss of an internet protocol packet.

Web page load times. Web page load times are defined as the time from submission of an internet URL into a web browser and the entire display of the requested web page. Load times are measured in milliseconds (ms).

Upload sustained trimmed mean (Mbps) peak weighted mean. This is a measure of download speed measured in megabits per second at peak hours (FCC OET, n.d.).

Upload sustained trimmed mean unweighted mean of percentage advertised. This is the ratio of the download speed compared to the advertised upload speed converted to percentage.

Appendix: Areas of Further Interest

Further study into reliability is needed to determine whether some reliability variables may have any impact. The reliability variables used in this study were insignificant either due to true insignificance or limitations of the data. Service availability as a measure of up time versus down time was not studied. The FCC does provide outage data in its NORS database, but this data is not broken down by the company reported or at least not publicly. Maintenance variables were not examined. Most companies do not provide items like Mean Time Between Failure (MTBF) or Mean Time to Repair (MTTR) (ITU, 2001 & 2008; ETSI 1994). Most companies do not provide maintenance or support as a separate line item in filed 10-Ks. This limitation was understood as a result that these variables would be strong detractors of service. These values being present would provide next-level transparency in the quality of service. Further literature review and analysis on maintenance variables and the impact on a company's bottom line is needed. Customer experience variables chosen in this study did not demonstrate a significant impact. Additional variables and literature can be found in Quality of Service versus Experience identified as QoS/E by ETSI (1994).

References

ACSI LLC. (n.d.). *Why ACSI*. Retrieved August 20, 2025, from https://theacsi.org/the-acsi-difference/why-acsi/

American Customer Satisfaction Index LLC (ACSI LLC). (2018, May 22). Press release — Telecommunications 2018 [Press release]. Retrieved August 20, 2025, from https://theacsi.org/news-and-resources/press-releases/2018/05/22/press-release-telecommunications-2018/

American Customer Satisfaction Index LLC (ACSI LLC). (2020, June 9). *ACSI telecommunications report 2019–2020* [Press release]. Retrieved August 20, 2025, from https://theacsi.org/news-and-resources/reports/2020/06/09/acsi-telecommunications-report-2019-2020/

American Customer Satisfaction Index LLC (ACSI LLC). (2021, June 8). Press release — Telecommunications study 2020–2021 [Press release]. Retrieved August 20, 2025, from https://theacsi.org/news-and-resources/press-releases/2021/06/08/press-release-telecommunications-2020-2021/

American Customer Satisfaction Index LLC (ACSI LLC). (2022, June 7). ACSI telecommunications study 2021–2022 [Press release]. Retrieved August 20, 2025, from https://theacsi.org/news-and-resources/reports/2022/06/07/acsi-telecommunications-study-2021-2022/

American Customer Satisfaction Index LLC (ACSI LLC). (2023, June 6). Press release — Telecommunications Study 2022–2023 [Press release]. Retrieved August 20, 2025, from https://theacsi.org/news-and-resources/press-releases/2023/06/06/press-release-telecommunications-study-2022-2023/

Altice USA, Inc. (2019). Form 10-K: Annual report for the fiscal year ended December 31, 2018. U.S. Securities and Exchange Commission.

https://www.sec.gov/Archives/edgar/data/1702780/000162828019002370/a2018123110-kalticeusa.htm

Altice USA, Inc. (2020). Form 10-K: Annual report for the fiscal year ended December 31, 2019. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001702780/000162828020001613/atus-20191231.htm

Altice USA, Inc. (2021). Form 10-K: Annual report for the fiscal year ended December 31, 2020. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001702780/000162828021001975/atus-20201231.htm

Altice USA, Inc. (2022). Form 10-K: Annual report for the fiscal year ended December 31, 2021. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001702780/000162828022002873/atus-20211231.htm

Altice USA, Inc. (2023). Form 10-K: Annual report for the fiscal year ended December 31, 2022. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001702780/000162828023004614/atus-20221231.htm

Altice USA, Inc. (2025). Form 10-K: Annual report for the fiscal year ended December 31, 2024. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001702780/000162828025005486/atus-20241231.htm

AT&T Inc. (2023). Form 10-K: Annual report for the fiscal year ended December 31, 2022. U.S. Securities and Exchange Commission.

https://www.sec.gov/Archives/edgar/data/732717/000073271723000011/t-20221231.htm

Beers, B. (2025, January 8). *Telecommunications sector: What it is and how to invest in it*. Investopedia. Reviewed by J. R. Brown; fact-checked by K. Rohrs Schmitt. Retrieved from https://www.investopedia.com/ask/answers/070815/what-telecommunications-sector.asp

CenturyLink, Inc. (2020). Form 10-K: Annual report for the fiscal year ended December 31, 2019. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000018926/000001892620000009/ctl2019123110k.htm

CenturyLink, Inc. (2021). Form 10-K: Annual report for the fiscal year ended December 31, 2020. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000018926/000001892620000009/ctl2019123110k.htm

Charter Communications, Inc. (2023). Form 10-K: Annual report for the fiscal year ended December 31, 2022. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001091667/000109166723000024/chtr-20221231.htm

Charter Communications, Inc. (2019). Form 10-K: Annual report for the fiscal year ended December 31, 2018. U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/1091667/000109166719000029/chtr12312018-10k.htm

Charter Communications, Inc. (2020). Form 10-K: Annual report for the fiscal year ended December 31, 2019. U.S. Securities and Exchange Commission.

https://www.sec.gov/ix?doc=/Archives/edgar/data/0001091667/000109166720000024/chtr12312019-10k.htm

Charter Communications, Inc. (2021). Form 10-K: Annual report for the fiscal year ended December 31, 2020. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001091667/000109166721000022/chtr-20201231.htm

Charter Communications, Inc. (2022). Form 10-K: Annual report for the fiscal year ended December 31, 2021. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001091667/000109166722000024/chtr-20211231.htm

Comcast Corporation. (2019). Form 10-K: Annual report for the fiscal year ended December 31, 2018. U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/1166691/000116669119000005/cmcsa-12312018x10k.htm

Comcast Corporation. (2020). Form 10-K: Annual report for the fiscal year ended December 31, 2019. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001166691/000116669120000008/cmcsa-12312019x10k.htm

Comcast Corporation. (2021). Form 10-K: Annual report for the fiscal year ended December 31, 2020. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001166691/000116669121000008/cmcsa-20201231.htm

Comcast Corporation. (2022). Form 10-K: Annual report for the fiscal year ended December 31, 2021. U.S. Securities and Exchange Commission.

https://www.sec.gov/ix?doc=/Archives/edgar/data/0001166691/000116669122000009/cmcsa-20211231.htm

Comcast Corporation. (2023). Form 10-K: Annual report for the fiscal year ended December 31, 2022. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001166691/000116669123000010/cmcsa-20221231.htm

European Telecommunications Standards Institute. (1994). Network aspects (NA); General aspects of quality of service (QoS) and network performance (NP) (ETR 003, 2nd ed.). https://www.etsi.org/deliver/etsi_etr/001_099/003/02_60/etr_003e02p.pdf

Federal Communications Commission, Office of Engineering and Technology. (2019). Statistical averages (September 2018) [Data set]. Measuring Broadband America. Retrieved August 19, 2025, from http://data.fcc.gov/download/measuring-broadband-america/2019/Statistical-Averages-sept2018.xlsx

Federal Communications Commission, Office of Engineering and Technology (FCC OET). (2020). Statistical averages (September 2019) [Data set]. Measuring Broadband America. Retrieved August 19, 2025, from http://data.fcc.gov/download/measuring-broadband-america/2020/Statistical-Averages-sept2019.xlsx

Federal Communications Commission, Office of Engineering and Technology (FCC OET). (2021). Statistical averages (September 2020) [Data set]. Measuring Broadband America. Retrieved August 19, 2025, from http://data.fcc.gov/download/measuring-broadband-america/2021/Statistical-Averages-sept2020.xlsx

Federal Communications Commission, Office of Engineering and Technology (FCC OET). (2022). Data processing flow (September 2021) [PDF]. Measuring Broadband America. Retrieved August 19, 2025, from https://data.fcc.gov/download/measuring-broadband-america/2022/data-processing-flow-sept2021.pdf

Federal Communications Commission, Office of Engineering and Technology (FCC OET). (2023). Statistical averages (September 2022) [Data set]. Measuring Broadband America. Retrieved August 19, 2025, from https://data.fcc.gov/download/measuring-broadband-america/2023/Statistical-Averages-sept2022.xlsx

Federal Communications Commission, Office of Engineering and Technology (FCC OET). (n.d.). Measuring Broadband America: Open methodology. Retrieved August 19, 2025, from https://www.fcc.gov/oet/mba/measuring-broadband-america-open-methodology

Frontier Communications Parent, Inc. (2023). Form 10-K: Annual report for the fiscal year ended December 31, 2022. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000020520/000156276223000055/fybr-20221231x10k.htm

Frontier Communications Corporation. (2019). Form 10-K: Annual report for the fiscal year ended December 31, 2018. U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/20520/000002052019000005/ftr-20181231x10k.htm

Frontier Communications Corporation. (2020). Form 10-K/A: Amendment to annual report for the fiscal year ended December 31, 2019. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000020520/000114036120009993/form10ka.htm

Frontier Communications Corporation. (2021). Form 10-K/A: Amendment to annual report for the fiscal year ended December 31, 2020. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000020520/000114036121007227/brhc10021007_10ka.htm

Frontier Communications Parent, Inc. (2022). Form 10-K: Annual report for the fiscal year ended December 31, 2021. U.S. Securities and Exchange Commission.

https://www.sec.gov/ix?doc=/Archives/edgar/data/0000020520/000000205202200000 4/fybr-20211231x10k.htm

International Telecommunication Union (ITU). (2001). Communications quality of service: A framework and definitions (Recommendation ITU-T G.1000). https://www.itu.int/rec/T-REC-G.1000-200111-I/en

International Telecommunication Union (ITU). (2008). Definitions of terms related to quality of service (Recommendation ITU-T E.800). https://www.itu.int/rec/T-REC-E.800-200809-I/en

International Telecommunication Union (ITU). (2017). Vocabulary for performance, quality of service and quality of experience (Recommendation ITU-T P.10/G.100). https://www.itu.int/rec/T-REC-P.10-201711-I/en

International Telecommunication Union, Telecommunication Development Bureau (ITU TDB). (2020). *ITU methodology for collecting administrative data for QoS indicators* [PDF]. Retrieved August 20, 2025, from https://www.itu.int/en/ITU-D/Statistics/Documents/publications/handbook/QoS_2020_indicators_methodology_web.pdf

International Telecommunication Union, Telecommunication Standardization Sector (ITU TSS). (1996, August 27). *Definitions and terminology for synchronization networks (ITU-T Rec. G.810 (08/96)).* ITU. Retrieved August 20, 2025, from https://www.itu.int/rec/dologin_pub.asp?lang=e&id=T-REC-G.810-199608-I!!PDF-E&type=items

Lumen Technologies, Inc. (2022). Form 10-K: Annual report for the fiscal year ended December 31, 2021. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000018926/000001892622000007/lumn-20211231.htm

Lumen Technologies, Inc. (2023). Form 10-K: Annual report for the fiscal year ended December 31, 2022. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000018926/000001892622000007/lumn-20211231.htm

Lumen Technologies, Inc. (2024). Form 10-K: Annual report for the fiscal year ended December 31, 2023. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000018926/000001892624000016/lumn-20231231.htm

Verizon Communications Inc. (2019). Form 10-K: Annual report for the fiscal year ended December 31, 2018. U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/732712/000073271219000012/a2018q410-k.htm

Verizon Communications Inc. (2020). Form 10-K: Annual report for the fiscal year ended December 31, 2019. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000732712/000073271220000014/a2019q410-k.htm

Verizon Communications Inc. (2021). Form 10-K: Annual report for the fiscal year ended December 31, 2020. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000732712/000073271221000012/vz-20201231.htm

Verizon Communications Inc. (2022). Form 10-K: Annual report for the fiscal year ended December 31, 2021. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000732712/000073271222000008/vz-20211231.htm

Verizon Communications Inc. (2023). Form 10-K: Annual report for the fiscal year ended December 31, 2022. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000732712/000073271223000012/vz-20221231.htm

Wierowski, M. (2025, March 3). *The state of competition in telecoms: Five commercial imperatives to regain an edge*. PwC. Retrieved from https://www.pwc.com/gx/en/industries/tmt/telecommunications/telco-state-of-competition.html